

Music Era and Release-Market Classification

DeepMIR 2026 Fall — Homework 1

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Code, checkpoints, README: <https://github.com/chijenpeng/26-fall-DeepMIR-HW1>

October 5, 2026

Experimental Setting

Pipeline

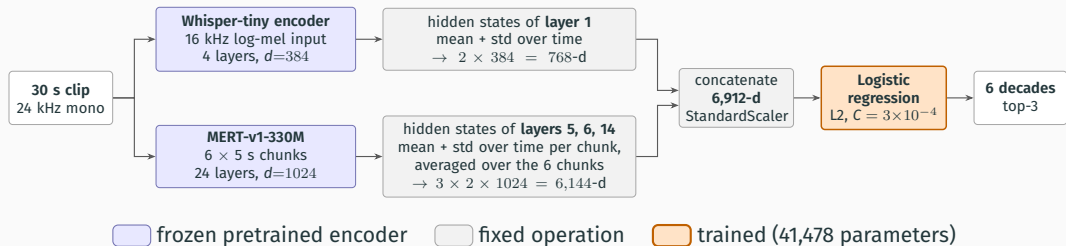
- 30 s clip, 24 kHz mono
- frozen encoder → mean + std pooling over time
- StandardScaler → L2 logistic regression
- train = fit, validation = model selection, test = prediction file only

Hardware / software

GPU	2 × NVIDIA RTX A4500 (20 GB)
CPU / RAM	Intel i9-14900K, 125 GB
OS	Ubuntu 26.04
Software	PyTorch 2.14, Transformers 4.45, scikit-learn 1.9

Results

Task 1: model



- **Whisper-tiny, layer 1:** shallow layers work best for decade; accuracy drops with depth
- **MERT-v1-330M, layers 5, 6, 14:** best three of a 25-layer sweep; adding Whisper-tiny gives +2.3 points
- **mean + std pooling:** one d -dim vector per frame $\rightarrow$ mean and std **over time** $\rightarrow 2 \times d$ values per layer
- MERT runs on 5 s chunks (its training length): statistics are taken inside each chunk, then averaged over the six
- Only the classifier is trained, with strong L2 regularization (1,026 clips, 6,912 features)

Task 1: release decade

Model: frozen encoders + logistic regression

- Whisper-tiny encoder, **layer 1**
- MERT-v1-330M, **layers 5, 6, 14**
- mean + std pooling over time (6,912-d)

Model and layers **chosen by ablation:** [▶ Section 3](#)

Validation, 132 clips	Top-1	Top-3
Best model	52.3%	86.4%

Errors

- **57%** of errors land on a **neighboring decade**
- Hardest: 2000s (recall .23), absorbed by 2010s and 1990s
- Easiest: 1960s and 2010s (one neighbor each)

True	1960s	17	2	0	3	0	0
	1970s	5	9	3	2	2	1
	1980s	2	2	13	2	3	0
	1990s	1	3	2	9	4	3
	2000s	2	2	1	5	5	7
	2010s	0	2	0	0	4	16
		1960s	1970s	1980s	1990s	2000s	2010s
		Predicted					

Counts. Rows = true decade (22 clips each).

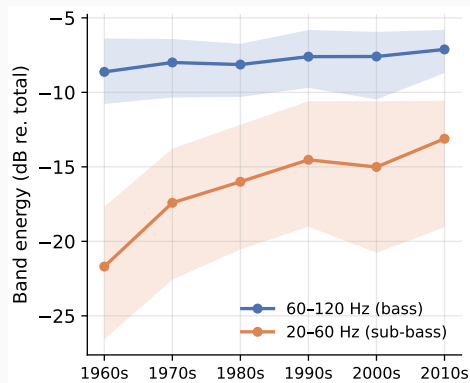
Why neighbors get confused (1): the low end

Hypothesis. Old masters were cut for vinyl, where deep bass makes the stylus jump, so they should carry less sub-bass.

Analysis. Long-term average spectrum (Welch) of the 1,026 training clips, split into two bands.

dB re. total	60s	70s	80s	90s	00s	10s
20-60 Hz	-21.7	-17.4	-16.0	-14.5	-15.0	-13.1
60-120 Hz	-8.6	-8.0	-8.1	-7.6	-7.6	-7.1

- Sub-bass rises by 8.6 dB from the 1960s to the 2010s; the 60-120 Hz band barely moves
- From the 1990s on the low end is nearly identical, so it cannot separate 1990s / 2000s / 2010s



Median and inter-quartile range, training split.

Why neighbors get confused (2): the loudness war

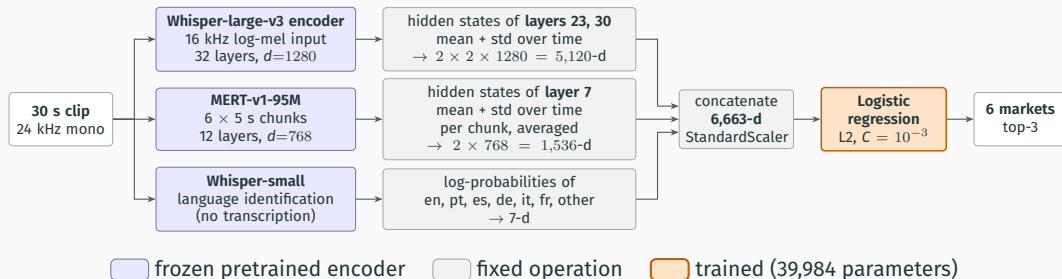
Hypothesis. From the 2000s, brick-wall limiters make masters louder and flatten peaks.

Median per decade, training clips	60s	70s	80s	90s	00s	10s
RMS level (dBFS)	-14.0	-14.6	-14.5	-13.0	-11.0	-10.6
Crest factor, peak – RMS (dB)	13.9	14.5	14.5	13.0	11.0	10.6
Dynamic range, loudest 20% blocks (dB)	10.6	11.2	11.3	10.7	8.9	8.7
Longest run pinned at the peak (samples)	5	5	6	9	22	26
Samples within 0.5 dB of the peak (%)	0.02	0.01	0.02	0.06	0.42	0.50

- 2000s and 2010s clips are 3 dB louder, have 3 dB less crest, and hold peaks at the ceiling. The two are **almost indistinguishable**, matching the 2000s ↔ 2010s confusion
- Production cues only split the data into “old” and “new”; finer distinctions must come from musical content

Caveat. The clips are 24 kHz mono, decoded from lossy Opus (YouTube). Resampling, downmixing and lossy coding reshape waveform peaks, so these numbers are trends (ANOVA $p < 10^{-30}$), not exact properties of the original masters. The two sample-level rows are the most affected.

Task 2: model



- **Whisper-large-v3, layers 23, 30:** deepest layers work best, the opposite of Task 1; this fits language being the main cue
- **MERT-v1-95M, layer 7:** +2 points on top of Whisper (57.8% $\rightarrow$ 59.8%); a music encoder adds what a speech encoder misses
- **Whisper-small language ID:** an explicit 7-d posterior; a small gain (top-3 89.2% $\rightarrow$ 90.2%) because deep Whisper layers already encode language
- Pooling as in Task 1: mean and std over time $\rightarrow 2 \times d$ values per layer

Task 2: release market

Model: frozen encoders + logistic regression

- Whisper-large-v3 encoder, **layers 23, 30**
- MERT-v1-95M, **layer 7**
- Whisper-small language-ID posteriors

Model and layers **chosen by ablation:** [▶ Section 3](#)

Validation, 102 clips	Top-1	Top-3
Best model	60.8%	90.2%

Errors

- **US ↔ UK** is the largest confusion (9 clips)
- **UK is over-predicted:** 23 called UK, only 9 correct
- Brazil (.88) and Spain (.82) are rarely missed
- Germany, Italy: 12 of their 21 errors go to US / UK

True	US	11	4	0	0	1	1
	UK	5	9	0	0	2	1
	Brazil	0	0	15	0	0	2
	Spain	0	2	0	14	1	0
	Germany	3	4	1	2	6	1
	Italy	1	4	0	2	3	7
		US	UK	Brazil	Spain	Germany	Italy
		Predicted					

Counts. Rows = true market (17 clips each).

Why English-language markets get confused: language

The dataset has **no language label**. We estimate it with Whisper-small **language identification** on the 798 training clips, on the mixture and on Demucs-separated vocal stems.

Market	Home language	Home mixture	Detected as English	
			mixture	vocals
US	English	92%	–	–
UK	English	100%	–	–
Brazil	Portuguese	87%	11%	12%
Spain	Spanish	57%	39%	41%
Germany	German	17%	83%	82%
Italy	Italian	41%	55%	56%

Share of training clips whose most likely language (among en, pt, es, de, it) is the home language or English.

- Most German releases and over half of the Italian ones are **sung in English**, so their vocals carry no market cue
- Vocal stems give the same answer: the accompaniment is not what misleads the detector
- Language separates Brazil and Spain. The English-language group has to be split by genre and production style, a much weaker cue
- A release market is where the record was issued, not the artist's nationality or language

Optional Experiments

Input length: 5 / 10 / 15 / 30 s

Ablation

Frozen features vs. fine-tuning

Audio Language Model

Disclosure

References
